

HEY ROSARIO!

Barangay Sto. Rosario Digital Platform

"Keng Sto. Rosario, ating mantabe kekayu"

PHASE 4: DATABASE DESIGN

Entity-Relationship Diagram | SQL Schema | Sample Test Data

Barangay Sto. Rosario, Angeles City, Pampanga, Philippines
Community Digital Governance Platform

STEP 10: Entity-Relationship Diagram (ERD)

10.1 Database Planning Overview

The Hey Rosario! database was designed to support all seven major feature modules of the platform. The design follows Third Normal Form (3NF) to eliminate data redundancy, ensure referential integrity through foreign key constraints, and optimize query performance through strategic indexing. The database uses MySQL with the InnoDB storage engine, enabling full foreign key enforcement and ACID-compliant transactions.

The schema consists of 17 tables organized into five functional groups, described below.

Module Group	Tables Included	Purpose
User Management	users, password_resets, session_logs	Authentication, accounts, security audit
Government Services	permit_applications, permit_documents, application_logs	Permit requests, uploads, status tracking
Citizen Reporting	reports, report_photos, report_updates	Dynamic Feature 1: issue reporting workflow
Announcements	announcements, announcement_views	Dynamic Feature 2: alerts and news feed
Community Information	emergency_contacts, evacuation_centers, tourism_listings, tourism_photos, transparency_documents, project_records	Emergency, tourism, transparency modules

10.2 Entity-Relationship Diagram

Each table is shown with its fields, data types, key types (PK / FK), constraints, and descriptions.

Group 1 — User Management Tables

TABLE: users				
Key	Field Name	Data Type	Constraints	Description
PK	user_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique user identifier
	full_name	VARCHAR(150)	NOT NULL	Full legal name
	email	VARCHAR(255)	NOT NULL, UNIQUE	Login email, must be unique
	password_hash	VARCHAR(255)	NOT NULL	bcrypt hash via password_hash()

	role	ENUM	NOT NULL, DEFAULT 'citizen'	admin staff citizen
	verification_status	ENUM	NOT NULL, DEFAULT 'pending'	pending verified rejected suspended
	contact_number	VARCHAR(20)	DEFAULT NULL	Philippine mobile/landline
	address	TEXT	DEFAULT NULL	Residential address
	id_document_path	VARCHAR(500)	DEFAULT NULL	Path to uploaded gov-issued ID
	profile_photo_path	VARCHAR(500)	DEFAULT NULL	Optional profile picture path
	last_login	DATETIME	DEFAULT NULL	Timestamp of most recent login
	is_active	TINYINT(1)	NOT NULL, DEFAULT 1	0 = suspended/deactivated
	registered_at	DATETIME	NOT NULL, DEFAULT NOW()	Account creation timestamp
	updated_at	DATETIME	NOT NULL, ON UPDATE CURRENT_TIMESTAMP	Last profile update

TABLE: password_resets

Key	Field Name	Data Type	Constraints	Description
PK	reset_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique reset record
FK	user_id	INT UNSIGNED	NOT NULL → users.user_id	Account requesting reset
	reset_token	VARCHAR(255)	NOT NULL, UNIQUE	SHA-256 hashed token
	expires_at	DATETIME	NOT NULL	Token expiry (1 hour)
	used_at	DATETIME	DEFAULT NULL	NULL if not yet used
	created_at	DATETIME	NOT NULL, DEFAULT NOW()	Request timestamp

TABLE: session_logs

Key	Field Name	Data Type	Constraints	Description
PK	log_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique log entry
FK	user_id	INT UNSIGNED	NOT NULL → users.user_id	User who triggered event
	ip_address	VARCHAR(45)	DEFAULT NULL	IPv4 or IPv6
	user_agent	VARCHAR(500)	DEFAULT NULL	Browser/device string
	action	ENUM	NOT NULL, DEFAULT 'login'	login logout timeout
	created_at	DATETIME	NOT NULL, DEFAULT NOW()	Event timestamp

Group 2 — Government Services Tables

TABLE: permit_applications				
Key	Field Name	Data Type	Constraints	Description
PK	application_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique application ID
FK	user_id	INT UNSIGNED	NOT NULL → users.user_id	Citizen submitting the application
	permit_type	ENUM	NOT NULL	business_permit barangay_clearance etc.
	applicant_name	VARCHAR(150)	NOT NULL	Name on the application
	purpose	TEXT	NOT NULL	Stated reason for the permit
	status	ENUM	NOT NULL, DEFAULT 'submitted'	draft submitted under_review approved rejected released
	rejection_reason	TEXT	DEFAULT NULL	Reason if rejected
	submission_date	DATETIME	NOT NULL, DEFAULT NOW()	When submitted
	processing_date	DATETIME	DEFAULT NULL	When reviewed by staff
	release_date	DATETIME	DEFAULT NULL	When document was released
FK	processed_by	INT UNSIGNED	DEFAULT NULL → users.user_id	Admin/staff who processed it
	notes	TEXT	DEFAULT NULL	Internal processing notes
	updated_at	DATETIME	NOT NULL, ON UPDATE NOW()	Last status update

TABLE: permit_documents				
Key	Field Name	Data Type	Constraints	Description
PK	doc_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique document ID
FK	application_id	INT UNSIGNED	NOT NULL → permit_applications	Parent application (CASCADE DELETE)
	document_label	VARCHAR(150)	NOT NULL	e.g. 'Valid ID', 'DTI Certificate'
	file_path	VARCHAR(500)	NOT NULL	Secure path outside web root
	file_name_original	VARCHAR(255)	NOT NULL	Original filename from upload
	file_size_bytes	INT UNSIGNED	NOT NULL	File size for validation display
	mime_type	VARCHAR(100)	NOT NULL	e.g. application/pdf

	uploaded_at	DATETIME	NOT NULL, DEFAULT NOW()	Upload timestamp
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TABLE: application_logs				
Key	Field Name	Data Type	Constraints	Description
PK	log_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique log entry
FK	application_id	INT UNSIGNED	NOT NULL → permit_applications	Linked application (CASCADE DELETE)
FK	admin_id	INT UNSIGNED	NOT NULL → users.user_id	Staff member who made the change
	action	VARCHAR(100)	NOT NULL	Human-readable action description
	old_status	VARCHAR(50)	DEFAULT NULL	Status before change
	new_status	VARCHAR(50)	DEFAULT NULL	Status after change
	notes	TEXT	DEFAULT NULL	Additional context
	created_at	DATETIME	NOT NULL, DEFAULT NOW()	When change occurred

Group 3 — Citizen Reporting Tables (Dynamic Feature 1)

TABLE: reports				
Key	Field Name	Data Type	Constraints	Description
PK	report_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique report identifier
FK	user_id	INT UNSIGNED	NOT NULL → users.user_id	Citizen who filed the report
	category	ENUM	NOT NULL	streetlight sanitation road_hazard drainage noise_complaint other
	title	VARCHAR(200)	NOT NULL	Short descriptive title
	description	TEXT	NOT NULL	Detailed issue description
	location_address	VARCHAR(500)	NOT NULL	Text address of the problem
	status	ENUM	NOT NULL, DEFAULT 'pending'	pending in_progress resolved rejected
FK	assigned_to	INT UNSIGNED	DEFAULT NULL → users.user_id	Staff assigned to handle report
	date_submitted	DATETIME	NOT NULL, DEFAULT NOW()	Submission timestamp
	date_updated	DATETIME	NOT NULL, ON UPDATE NOW()	Last status change timestamp
	resolution_notes	TEXT	DEFAULT NULL	Final resolution description

TABLE: report_photos				
Key	Field Name	Data Type	Constraints	Description
PK	photo_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique photo ID
FK	report_id	INT UNSIGNED	NOT NULL → reports.report_id	Parent report (CASCADE DELETE)
	file_path	VARCHAR(500)	NOT NULL	Secure storage path
	file_name_original	VARCHAR(255)	NOT NULL	Original filename from upload
	file_size_bytes	INT UNSIGNED	NOT NULL	File size in bytes (max 5MB)
	is_primary	TINYINT(1)	NOT NULL, DEFAULT 0	1 = thumbnail in list views
	uploaded_at	DATETIME	NOT NULL, DEFAULT NOW()	Upload timestamp

TABLE: report_updates				
Key	Field Name	Data Type	Constraints	Description
PK	update_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique update record
FK	report_id	INT UNSIGNED	NOT NULL → reports.report_id	Linked report (CASCADE DELETE)
FK	admin_id	INT UNSIGNED	NOT NULL → users.user_id	Staff who made the update (RESTRICT)
	old_status	ENUM	DEFAULT NULL	Status before change
	new_status	ENUM	NOT NULL	Status after change
	comment	TEXT	DEFAULT NULL	Staff note explaining the update
	created_at	DATETIME	NOT NULL, DEFAULT NOW()	When update was made

Group 4 — Announcements Tables (Dynamic Feature 2)

TABLE: announcements				
Key	Field Name	Data Type	Constraints	Description
PK	announcement_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique announcement ID
	title	VARCHAR(255)	NOT NULL	Announcement headline
	content	LONGTEXT	NOT NULL	Rich HTML content (sanitized)
	category	ENUM	NOT NULL, DEFAULT 'general'	general emergency events projects health weather
	priority	TINYINT UNSIGNED	NOT NULL, DEFAULT 1	1=Normal, 2=High, 3=Emergency
FK	author_id	INT UNSIGNED	NOT NULL → users.user_id	Admin/staff who created it
	cover_image_path	VARCHAR(500)	DEFAULT NULL	Optional header image path
	status	ENUM	NOT NULL, DEFAULT 'draft'	draft published archived
	date_posted	DATETIME	DEFAULT NULL	NULL until published
	date_archived	DATETIME	DEFAULT NULL	When archived (if applicable)
	created_at	DATETIME	NOT NULL, DEFAULT NOW()	Record creation time
	updated_at	DATETIME	NOT NULL, ON UPDATE NOW()	Last edit timestamp

TABLE: announcement_views				
Key	Field Name	Data Type	Constraints	Description
PK	view_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique view record
FK	announcement_id	INT UNSIGNED	NOT NULL → announcements	Viewed announcement (CASCADE DELETE)
FK	user_id	INT UNSIGNED	NOT NULL → users.user_id	User who viewed it (CASCADE DELETE)
	viewed_at	DATETIME	NOT NULL, DEFAULT NOW()	View timestamp
	(UNIQUE)	(announcement_id + user_id)	UNIQUE KEY	One record per user per announcement

Group 5 — Community Information Tables

TABLE: emergency_contacts				
Key	Field Name	Data Type	Constraints	Description
PK	contact_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique contact ID
	category	ENUM	NOT NULL	barangay police fire medical rescue utility
	org_name	VARCHAR(200)	NOT NULL	Organization or office name
	primary_phone	VARCHAR(30)	NOT NULL	Main emergency contact number
	secondary_phone	VARCHAR(30)	DEFAULT NULL	Alternate number
	address	VARCHAR(500)	DEFAULT NULL	Physical location
	operating_hours	VARCHAR(150)	DEFAULT NULL	e.g. '24/7' or '8AM-5PM'
	sort_order	TINYINT UNSIGNED	NOT NULL, DEFAULT 99	Display order within category
	is_active	TINYINT(1)	NOT NULL, DEFAULT 1	0 = hidden from directory

TABLE: evacuation_centers				
Key	Field Name	Data Type	Constraints	Description
PK	center_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique center ID
	name	VARCHAR(200)	NOT NULL	Center name
	address	VARCHAR(500)	NOT NULL	Physical address
	capacity_persons	INT UNSIGNED	DEFAULT NULL	Maximum evacuee capacity
	status	ENUM	NOT NULL, DEFAULT 'standby'	standby active full closed
	contact_person	VARCHAR(150)	DEFAULT NULL	In-charge person
	contact_phone	VARCHAR(30)	DEFAULT NULL	Contact number at center
	facilities	TEXT	DEFAULT NULL	Available facilities list
	notes	TEXT	DEFAULT NULL	Additional information

TABLE: tourism_listings				
Key	Field Name	Data Type	Constraints	Description
PK	listing_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique listing ID
	name	VARCHAR(200)	NOT NULL	Establishment or attraction name
	category	ENUM	NOT NULL	cultural food business recreation religious
	description	TEXT	NOT NULL	Full description
	address	VARCHAR(500)	NOT NULL	Location address

	operating_hours	VARCHAR(150)	DEFAULT NULL	Business hours
	contact_number	VARCHAR(30)	DEFAULT NULL	Phone/mobile contact
	facebook_url	VARCHAR(500)	DEFAULT NULL	Optional Facebook page
	status	ENUM	NOT NULL, DEFAULT 'active'	active inactive (soft delete)
FK	added_by	INT UNSIGNED	NOT NULL → users.user_id	Admin who added the listing

TABLE: tourism_photos				
Key	Field Name	Data Type	Constraints	Description
PK	photo_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique photo ID
FK	listing_id	INT UNSIGNED	NOT NULL → tourism_listings	Parent listing (CASCADE DELETE)
	file_path	VARCHAR(500)	NOT NULL	Secure file storage path
	caption	VARCHAR(300)	DEFAULT NULL	Photo caption text
	is_primary	TINYINT(1)	NOT NULL, DEFAULT 0	1 = cover/card thumbnail
	sort_order	TINYINT UNSIGNED	NOT NULL, DEFAULT 99	Gallery display order
	uploaded_at	DATETIME	NOT NULL, DEFAULT NOW()	Upload timestamp

TABLE: transparency_documents				
Key	Field Name	Data Type	Constraints	Description
PK	doc_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique document ID
	title	VARCHAR(300)	NOT NULL	Official document title
	description	TEXT	DEFAULT NULL	Brief summary
	category	ENUM	NOT NULL	executive_order ordinance resolution financial_report project_record minutes other
	document_number	VARCHAR(100)	DEFAULT NULL	e.g. 'EO No. 2024-001'
	date_issued	DATE	DEFAULT NULL	Official document date
	file_path	VARCHAR(500)	NOT NULL	Secure path outside web root
	file_name_original	VARCHAR(255)	NOT NULL	Original uploaded filename
	file_size_bytes	INT UNSIGNED	NOT NULL	File size for display

	mime_type	VARCHAR (100)	NOT NULL	e.g. application/pdf
	download_count	INT UNSIGNED	NOT NULL, DEFAULT 0	Incremented on each download
	status	ENUM	NOT NULL, DEFAULT 'published'	published archived draft
FK	uploaded_by	INT UNSIGNED	NOT NULL → users.user_id	Admin who uploaded the document

TABLE: project_records				
Key	Field Name	Data Type	Constraints	Description
PK	project_id	INT UNSIGNED	NOT NULL, AUTO_INCREMENT	Unique project ID
	title	VARCHAR (300)	NOT NULL	Project title
	description	TEXT	NOT NULL	Full project description
	status	ENUM	NOT NULL, DEFAULT 'planned'	planned ongoing completed cancelled
	budget_allocated	DECIMAL (15,2)	DEFAULT NULL	Total approved budget (PHP)
	budget_spent	DECIMAL (15,2)	DEFAULT NULL	Actual expenditure (PHP)
	start_date	DATE	DEFAULT NULL	Project start date
	target_end_date	DATE	DEFAULT NULL	Planned completion date
	actual_end_date	DATE	DEFAULT NULL	Actual completion date
	implementing_office	VARCHAR (200)	DEFAULT NULL	Responsible office/department
	funding_source	VARCHAR (200)	DEFAULT NULL	Budget source (e.g. Dev Fund)
	location_area	VARCHAR (300)	DEFAULT NULL	Geographic area affected
FK	added_by	INT UNSIGNED	NOT NULL → users.user_id	Admin who added the record

10.3 Relationship Summary

The following table summarizes all foreign key relationships in the database, including the delete behavior enforced by InnoDB.

Parent Table	Child Table	Relationship	FK Column	On Delete
users	password_resets	One-to-Many	user_id	CASCADE
users	session_logs	One-to-Many	user_id	CASCADE
users	permit_applications	One-to-Many	user_id	RESTRICT
users	permit_applications	One-to-Many (processor)	processed_by	SET NULL
permit_applications	permit_documents	One-to-Many	application_id	CASCADE
permit_applications	application_logs	One-to-Many	application_id	CASCADE
users	application_logs	One-to-Many	admin_id	RESTRICT
users	reports	One-to-Many	user_id	RESTRICT
users	reports	One-to-Many (assignee)	assigned_to	SET NULL
reports	report_photos	One-to-Many	report_id	CASCADE
reports	report_updates	One-to-Many	report_id	CASCADE
users	report_updates	One-to-Many	admin_id	RESTRICT
users	announcements	One-to-Many	author_id	RESTRICT
announcements	announcement_views	One-to-Many	announcement_id	CASCADE
users	announcement_views	One-to-Many	user_id	CASCADE
users	tourism_listings	One-to-Many	added_by	RESTRICT
tourism_listings	tourism_photos	One-to-Many	listing_id	CASCADE
users	transparency_documents	One-to-Many	uploaded_by	RESTRICT
users	project_records	One-to-Many	added_by	RESTRICT

KEY LEGEND:

PK = Primary Key | FK = Foreign Key | CASCADE = child deleted when parent deleted |
 RESTRICT = parent cannot be deleted while children exist | SET NULL = FK column set to NULL when parent is deleted

STEP 11: SQL Schema — CREATE DATABASE & CREATE TABLE

11.1 Database Setup

The Hey Rosario! database is created using the following configuration. The utf8mb4 character set is required to properly support Filipino characters (such as accented vowels in Kapampangan text), emoji, and other special Unicode characters used in announcement content.

```
CREATE DATABASE IF NOT EXISTS hey_rosario_db
    CHARACTER SET utf8mb4
    COLLATE utf8mb4_unicode_ci;

USE hey_rosario_db;
```

11.2 Table Creation Logic & Design Decisions

Table 1–3: User Management

The users table is the anchor of the entire schema. Every other table that involves a person—whether as a citizen submitting a report, an admin approving a permit, or a staff member posting an announcement—references users.user_id as a foreign key. The password_hash column stores only a bcrypt hash generated by PHP's password_hash() function; the plaintext password is never persisted. The verification_status ENUM column controls access gating: citizens with 'pending' or 'rejected' status cannot submit service requests.

The session_logs table records every login, logout, and session timeout event. This supports both security auditing (detecting unauthorized access attempts) and usage analytics (understanding peak platform usage times).

```
CREATE TABLE users (
    user_id          INT UNSIGNED NOT NULL AUTO_INCREMENT,
    full_name        VARCHAR(150) NOT NULL,
    email            VARCHAR(255) NOT NULL,
    password_hash     VARCHAR(255) NOT NULL,
    role             ENUM('admin','staff','citizen') NOT NULL DEFAULT 'citizen',
    verification_status ENUM('pending','verified','rejected','suspended')
                    NOT NULL DEFAULT 'pending',
    contact_number    VARCHAR(20)  DEFAULT NULL,
    address           TEXT          DEFAULT NULL,
    id_document_path  VARCHAR(500) DEFAULT NULL,
    is_active         TINYINT(1)   NOT NULL DEFAULT 1,
    registered_at     DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at        DATETIME      NOT NULL DEFAULT CURRENT_TIMESTAMP
```

```

                                ON UPDATE CURRENT_TIMESTAMP,
PRIMARY KEY (user_id),
UNIQUE KEY uq_users_email (email),
INDEX idx_users_role (role),
INDEX idx_users_verification_status (verification_status)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tables 4–6: Government Services

The `permit_applications` table uses an ENUM for `permit_type` to constrain the accepted values to the defined set of barangay services. The 'draft' status allows citizens to save partially completed applications before final submission. The `processed_by` foreign key uses SET NULL on delete, ensuring that if an admin account is removed, existing application records retain their data (with `processed_by` set to NULL) rather than being deleted.

The `permit_documents` table separates uploaded files from the application record itself, allowing multiple documents per application without requiring complex JSON columns or denormalization. Each document file is stored outside the web root—the `file_path` column holds the server-side path, never a URL—and served via a PHP streaming script that validates session permissions.

The `application_logs` table provides a complete audit trail. Every time staff changes an application's status, a log entry is created recording who made the change, when, and what the old and new statuses were. This satisfies accountability requirements for government service records.

```

CREATE TABLE permit_applications (
  application_id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id        INT UNSIGNED NOT NULL,
  permit_type    ENUM('business_permit','barangay_clearance',
                      'certificate_of_residency','certificate_of_indigency',
                      'first_time_jobseeker','other') NOT NULL,
  applicant_name VARCHAR(150) NOT NULL,
  purpose        TEXT          NOT NULL,
  status         ENUM('draft','submitted','under_review',
                      'approved','rejected','released') NOT NULL DEFAULT
'submitted',
  rejection_reason TEXT          DEFAULT NULL,
  submission_date DATETIME       NOT NULL DEFAULT CURRENT_TIMESTAMP,
  processing_date  DATETIME       DEFAULT NULL,
  processed_by     INT UNSIGNED  DEFAULT NULL,
  PRIMARY KEY (application_id),
  INDEX idx_permit_status (status),
  INDEX idx_permit_type (permit_type),
  CONSTRAINT fk_permit_user
    FOREIGN KEY (user_id) REFERENCES users(user_id)
    ON DELETE RESTRICT ON UPDATE CASCADE,
  CONSTRAINT fk_permit_processor
    FOREIGN KEY (processed_by) REFERENCES users(user_id)
    ON DELETE SET NULL ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;

```

Tables 7–9: Citizen Reporting (Dynamic Feature 1)

The reports table uses ON DELETE RESTRICT for the user_id foreign key, preventing citizen accounts from being deleted while their reports remain in the system. This is intentional—even if a citizen is suspended or deactivated (is_active = 0), their reports must remain accessible to administrators for resolution tracking. The assigned_to column uses SET NULL, so reports are not deleted if the assigned staff member's account is removed.

The report_photos table uses CASCADE DELETE, meaning all photos attached to a report are automatically removed when the report itself is deleted. This prevents orphaned file records from accumulating in the database.

The report_updates table uses RESTRICT on admin_id, preventing deletion of admin accounts that have processed reports. This preserves the complete accountability trail—every status change has an attributed responsible person in the audit log.

```
CREATE TABLE reports (
  report_id      INT UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id        INT UNSIGNED NOT NULL,
  category       ENUM('streetlight','sanitation','road_hazard',
    'drainage','noise_complaint','other') NOT NULL,
  title          VARCHAR(200) NOT NULL,
  description     TEXT NOT NULL,
  location_address VARCHAR(500) NOT NULL,
  status         ENUM('pending','in_progress','resolved','rejected')
    NOT NULL DEFAULT 'pending',
  assigned_to     INT UNSIGNED DEFAULT NULL,
  date_submitted  DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  date_updated    DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
    ON UPDATE CURRENT_TIMESTAMP,
  PRIMARY KEY (report_id),
  INDEX idx_reports_category (category),
  INDEX idx_reports_status (status),
  CONSTRAINT fk_report_user
    FOREIGN KEY (user_id) REFERENCES users(user_id)
    ON DELETE RESTRICT ON UPDATE CASCADE,
  CONSTRAINT fk_report_assigned
    FOREIGN KEY (assigned_to) REFERENCES users(user_id)
    ON DELETE SET NULL ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Tables 10–11: Announcements (Dynamic Feature 2)

The announcements table includes both a category ENUM and a numeric priority column. While category describes the type of announcement (General, Emergency, Events, etc.), priority is a separate integer (1–3) used for display ordering: ORDER BY priority DESC, date_posted DESC ensures that priority-3 Emergency alerts always appear at the top of the feed regardless of posting

date. This separation allows, for example, a weather advisory to be categorized as 'weather' but given Emergency-level priority.

A FULLTEXT INDEX on (title, content) enables efficient keyword search queries using MySQL's MATCH...AGAINST syntax, which is significantly faster than LIKE '%keyword%' for large content tables.

```
CREATE TABLE announcements (
  announcement_id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  title           VARCHAR(255) NOT NULL,
  content         LONGTEXT    NOT NULL,
  category        ENUM('general','emergency','events',
    'projects','health','weather') NOT NULL DEFAULT 'general',
  priority        TINYINT UNSIGNED NOT NULL DEFAULT 1,
  author_id       INT UNSIGNED NOT NULL,
  status          ENUM('draft','published','archived') NOT NULL DEFAULT 'draft',
  date_posted     DATETIME     DEFAULT NULL,
  PRIMARY KEY (announcement_id),
  INDEX idx_announce_priority (priority),
  FULLTEXT INDEX ft_announce_search (title, content),
  CONSTRAINT fk_announce_author
    FOREIGN KEY (author_id) REFERENCES users(user_id)
    ON DELETE RESTRICT ON UPDATE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

Tables 12–17: Community Information

The emergency_contacts table has no foreign keys—it is entirely self-contained and admin-managed, suitable for content that changes infrequently and does not depend on user accounts. The sort_order column allows admins to control the display sequence of contacts within each category (e.g., ensuring the Barangay Hall main line appears before other barangay contacts).

The tourism_listings table uses a soft-delete pattern: instead of physically deleting records, the status column is set to 'inactive'. This preserves listing history and prevents broken references in analytics logs or cached pages. The FULLTEXT index supports keyword search across both name and description fields.

The transparency_documents table tracks download_count as a counter column, incremented by the application layer each time a document is downloaded. This supports admin analytics on which documents are most accessed by the public.

The project_records table includes both budget_allocated and budget_spent as DECIMAL(15,2) columns, supporting precise financial tracking in Philippine Pesos with two decimal places. The target_end_date and actual_end_date allow comparison for timeline adherence reporting.

STEP 12: Test Sample Data — INSERT INTO

12.1 Sample Data Overview

The following sample data was inserted to test all core system workflows: user registration and verification, permit application lifecycle, citizen report submission and resolution, announcement publishing, emergency contact directory, and transparency document management. The data set uses realistic Filipino names, Barangay Sto. Rosario addresses, and actual barangay services and concerns.

Table	Records Inserted	Coverage
users	10	2 admins, 2 staff, 4 verified citizens, 1 pending, 1 suspended
permit_applications	6	All status stages: submitted, under_review, approved, rejected, released
application_logs	6	Status change audit trail for key applications
reports	6	All categories: streetlight, road hazard, sanitation, drainage, noise
report_updates	6	Status transitions: pending → in_progress → resolved
announcements	8	All categories, both priority 1 and priority 3; 1 draft, 7 published
emergency_contacts	14	All 6 categories: barangay, police, fire, medical, rescue, utility
evacuation_centers	3	1 active, 2 standby
tourism_listings	7	All 5 categories: religious, cultural, food (3), business, recreation
transparency_documents	5	executive_order, ordinance, resolution, financial_report, minutes
project_records	5	All statuses: planned (2), ongoing (1), completed (2)

12.2 Sample INSERT Statements — Key Excerpts

Users Table (Sample)

The password_hash values shown below represent bcrypt hashes as they would be generated by PHP's password_hash(PASSWORD_BCRYPT). In a real deployment, these are never identical—bcrypt generates a unique salt each time even for the same plaintext password.

```
INSERT INTO users
```

```

        (user_id, full_name, email, password_hash, role,
         verification_status, contact_number, address)
VALUES
-- Admin
(1, 'Maria Santos Cruz', 'admin@heyrosario.gov.ph',
 '$2y$12$...bcrypt_hash...', 'admin', 'verified',
 '09171234567', 'Barangay Hall, Sto. Rosario, Angeles City'),

-- Verified Citizen
(5, 'Luzviminda Ocampo Torres', 'luz.ocampo@gmail.com',
 '$2y$12$...bcrypt_hash...', 'citizen', 'verified',
 '09501234567', '123 Mabini St., Sto. Rosario, Angeles City'),

-- Pending Citizen (awaiting ID verification)
(9, 'Maricel Ramos Dizon', 'maricel.ramos@gmail.com',
 '$2y$12$...bcrypt_hash...', 'citizen', 'pending',
 '09941234567', '34 Del Pilar St., Sto. Rosario, Angeles City');

```

Permit Applications Table (Sample)

The sample data covers all six possible status values to allow testing of every stage of the permit application workflow in the admin dashboard.

```

INSERT INTO permit_applications
    (application_id, user_id, permit_type, applicant_name,
     purpose, status, submission_date, processed_by)
VALUES
-- APPROVED: Business permit, fully processed
(1, 5, 'business_permit', 'Luzviminda Ocampo Torres',
 'Apply for new sari-sari store at 123 Mabini St.',
 'approved', '2025-01-05 09:00:00', 3),

-- UNDER REVIEW: Certificate of residency, still processing
(3, 7, 'certificate_of_residency', 'Teresita Villanueva Lopez',
 'Proof of residency for SSS benefit claim.',
 'under_review', '2025-01-10 11:30:00', NULL),

-- REJECTED: Business permit renewal, missing documents
(6, 7, 'business_permit', 'Teresita Villanueva Lopez',
 'Renewal of carinderia permit for 2025.',
 'rejected', '2025-01-08 14:00:00', 3);

```

Reports Table — Dynamic Feature 1 (Sample)

Report samples include all defined categories and all four status values. This allows testing of the admin report management dashboard's filter controls and the citizen's report history view.

```

INSERT INTO reports
  (report_id, user_id, category, title, description,
   location_address, status, assigned_to, resolution_notes)
VALUES
-- RESOLVED: Streetlight (with resolution note)
(1, 5, 'streetlight',
 'Broken Streetlight on Mabini St. Corner Rizal Ave.',
 'Non-functional for 3 days. Dark area at night, safety risk.',
 'Corner Mabini St. and Rizal Ave., Sto. Rosario',
 'resolved', 3,
 'Bulb replaced by maintenance team on Jan 8. '),

-- IN PROGRESS: Road hazard near school
(2, 6, 'road_hazard',
 'Large Pothole on Luna Street Near Elementary School',
 '1-meter pothole in front of school. Multiple motorcycles damaged.',
 'Luna St. in front of Sto. Rosario Elementary School',
 'in_progress', 4, NULL),

-- PENDING: Noise complaint (new, unassigned)
(6, 6, 'noise_complaint',
 'Excessive Noise from Videoke Bar Past Midnight',
 'Operating past midnight on weekends, disturbing residents.',
 'Rizal Ave. near Purok 4 entrance, Sto. Rosario',
 'pending', NULL, NULL);

```

Announcements Table — Dynamic Feature 2 (Sample)

The sample announcements include two Emergency-priority records (priority = 3) to test the top-of-feed placement and red banner display logic, one draft (unpublished), and records across all six category types.

```

INSERT INTO announcements
  (announcement_id, title, content, category,
   priority, author_id, status, date_posted)
VALUES
-- EMERGENCY, Priority 3: Typhoon advisory
(1, 'EMERGENCY ALERT: Typhoon Carina Preparation Advisory',
 '<p><strong>EMERGENCY ADVISORY</strong>...</p>',
 'emergency', 3, 1, 'published', '2025-01-08 06:00:00'),

-- WEATHER, Priority 3: Flood warning
(7, 'FLOOD WARNING: Heavy Rains Expected January 12-14',
 '<p><strong>WEATHER ADVISORY</strong>...</p>',
 'weather', 3, 1, 'published', '2025-01-12 05:30:00'),

-- GENERAL, Priority 1: Draft (not yet published)
(8, 'New Business Permit Application Process for 2025',
 '<p>The Barangay Business Permit Application...</p>',
 'general', 1, 1, 'draft', '2025-01-01 00:00:00');

```

```
'general', 1, 2, 'draft', NULL);
```

Emergency Contacts Table (Sample)

```
INSERT INTO emergency_contacts
    (category, org_name, primary_phone, secondary_phone,
     address, operating_hours, sort_order)
VALUES
    ('barangay', 'Barangay Sto. Rosario - Main Office',
     '(045) 888-0001', '0917-888-0001',
     'Barangay Hall, Sto. Rosario, Angeles City',
     '8:00 AM - 5:00 PM, Mon-Fri', 1),

    ('police', 'Angeles City Police Station (Main)',
     '(045) 322-4343', '117',
     'Nepomuceno St., Angeles City', '24/7', 1),

    ('medical', 'Angeles University Foundation Medical Center',
     '(045) 625-4100', '0917-AUF-HELP',
     'MacArthur Highway, Angeles City', '24/7 Emergency Room', 1);
```

Transparency Documents Table (Sample)

```
INSERT INTO transparency_documents
    (title, description, category, document_number, date_issued,
     file_path, file_name_original, file_size_bytes, mime_type,
     download_count, status, uploaded_by)
VALUES
    ('Executive Order No. 2024-001: Establishment of BDRRMC',
     'EO creating the official Barangay Disaster Risk Reduction Council.',
     'executive_order', 'EO No. 2024-001', '2024-02-15',
     '/docs/transparency/EO_2024_001.pdf',
     'EO_2024_001_Disaster_Council.pdf',
     1258291, 'application/pdf', 47, 'published', 1),

    ('Barangay Ordinance No. 2024-003: Solid Waste Management',
     'Ordinance mandating waste segregation with penalties.',
     'ordinance', 'BRO No. 2024-003', '2024-04-10',
     '/docs/transparency/ORD_2024_003.pdf',
     'ORD_2024_003_Solid_Waste.pdf',
     956432, 'application/pdf', 63, 'published', 1);
```

12.3 Key SQL Queries Used by the Application

Query 1 — Announcement Feed (Dynamic Feature 2)

Retrieves published announcements with priority-based and date-based ordering. Emergency alerts (priority=3) always appear first.

```
SELECT a.announcement_id, a.title, a.category, a.priority,
       a.content, a.date_posted, u.full_name AS author_name
FROM announcements a
JOIN users u ON a.author_id = u.user_id
WHERE a.status = 'published'
ORDER BY a.priority DESC, a.date_posted DESC
LIMIT 10 OFFSET ?; -- AJAX pagination
```

Query 2 — Announcement Keyword Search

Uses FULLTEXT index for fast keyword matching across title and content.

```
SELECT announcement_id, title, category, priority, date_posted
FROM announcements
WHERE status = 'published'
      AND MATCH(title, content) AGAINST(? IN BOOLEAN MODE)
ORDER BY priority DESC, date_posted DESC;
```

Query 3 — Citizen Report History with Primary Photo

Retrieves a citizen's reports with their primary thumbnail photo using a LEFT JOIN.

```
SELECT r.report_id, r.category, r.title, r.status,
       r.location_address, r.date_submitted,
       p.file_path AS primary_photo
FROM reports r
LEFT JOIN report_photos p
      ON p.report_id = r.report_id AND p.is_primary = 1
WHERE r.user_id = ?
ORDER BY r.date_submitted DESC;
```

Query 4 — Admin Report Dashboard with Filters

The admin report queue with optional category and status filtering, plus the name of the assigned staff member.

```
SELECT r.report_id, r.category, r.title, r.status,
       r.date_submitted, u.full_name AS citizen_name,
       a.full_name AS assigned_staff
FROM reports r
JOIN users u ON r.user_id = u.user_id
LEFT JOIN users a ON r.assigned_to = a.user_id
WHERE (? IS NULL OR r.category = ?)
      AND (? IS NULL OR r.status = ?)
ORDER BY r.date_submitted ASC;
```

Query 5 — Report Status Update (Transaction)

Status changes are wrapped in a transaction to ensure the report record and audit log entry are always updated together atomically.

```
START TRANSACTION;

UPDATE reports
SET status = ?, date_updated = NOW()
WHERE report_id = ?;

INSERT INTO report_updates
    (report_id, admin_id, old_status, new_status, comment)
VALUES (?, ?, ?, ?, ?);

COMMIT;
```

Query 6 — Permit Application Admin Queue

Shows submitted and under-review applications with citizen details, ordered oldest-first to prioritize longest-waiting applicants.

```
SELECT pa.application_id, pa.permit_type, pa.applicant_name,
       pa.status, pa.submission_date,
       u.email AS citizen_email, u.contact_number
FROM permit_applications pa
JOIN users u ON pa.user_id = u.user_id
WHERE pa.status IN ('submitted', 'under_review')
ORDER BY pa.submission_date ASC;
```

Query 7 — Transparency Portal Search

Searches official documents by keyword using FULLTEXT index, optionally filtered by category.

```
SELECT doc_id, title, category, document_number,
       date_issued, file_name_original, file_size_bytes,
       download_count
FROM transparency_documents
WHERE status = 'published'
      AND (? IS NULL OR category = ?)
      AND MATCH(title, description) AGAINST(? IN BOOLEAN MODE)
ORDER BY date_issued DESC;
```

Query 8 — Admin Statistics Dashboard

Aggregation queries used to populate the statistics widgets on the admin dashboard.

```
-- Active reports by status
```

```
SELECT status, COUNT(*) AS total
FROM reports
GROUP BY status;

-- Permit applications by status
SELECT status, COUNT(*) AS total
FROM permit_applications
GROUP BY status;

-- Pending citizen verifications
SELECT COUNT(*) AS pending_verifications
FROM users
WHERE verification_status = 'pending';

-- Most-downloaded transparency documents
SELECT title, download_count
FROM transparency_documents
ORDER BY download_count DESC
LIMIT 5;
```

12.4 Data Integrity Verification Queries

These SELECT queries can be run after inserting all sample data to verify that all relationships, statuses, and values are correct.

```
-- Verify all users have valid roles
SELECT role, COUNT(*) FROM users GROUP BY role;

-- Verify FK integrity: all reports reference valid users
SELECT COUNT(*) FROM reports r
LEFT JOIN users u ON r.user_id = u.user_id
WHERE u.user_id IS NULL; -- Should return 0

-- Check report_updates audit trail completeness
SELECT r.report_id, r.status AS current_status,
       MAX(ru.created_at) AS last_update
FROM reports r
LEFT JOIN report_updates ru ON r.report_id = ru.report_id
GROUP BY r.report_id, r.status;

-- Confirm emergency announcements appear first in feed
SELECT title, priority, category, date_posted
FROM announcements
WHERE status = 'published'
ORDER BY priority DESC, date_posted DESC
LIMIT 5;
```